arXiv:2505.19176v3 [cs.CL] 18 Sep 2025

Assistant-Guided Mitigation of Teacher Preference Bias in LLM-as-a-Judge

Zhuo Liu[1], **Moxin Li**[2,†], **Xun Deng**[1], **Qifan Wang**[3], **Fuli Feng**[1,†],
[1]University of Science and Technology of China [2]National University of Singapore [3]Meta AI,
liuz_@mail.ustc.edu.cn, limoxin@u.nus.edu, dx981228@mail.ustc.edu.cn,
wqfcr@fb.com, fulifeng93@gmail.com

Abstract

LLM-as-a-Judge employs large language models (LLMs), such as GPT-4, to evaluate the quality of LLM-generated responses, gaining popularity for its cost-effectiveness and strong alignment with human evaluations. However, training proxy judge models using evaluation data generated by powerful teacher models introduces a critical yet previously overlooked issue: teacher preference bias, where the proxy judge model learns a biased preference for responses from the teacher model. To tackle this problem, we propose a novel setting that incorporates an additional assistant model, which is not biased toward the teacher model's responses, to complement the training data. Building on this setup, we introduce AGDe-Judge, a three-stage framework designed to debias from both the labels and feedbacks in the training data. Extensive experiments demonstrate that AGDe-Judge effectively reduces teacher preference bias while maintaining strong performance across six evaluation benchmarks[1].

1 Introduction

LLM-as-a-Judge refers to the use of Large Language Models (LLMs), such as GPT-4 (OpenAI, 2024), to evaluate text quality by generating feedback and making evaluative judgments (Zheng et al., 2023). Unlike traditional manual evaluation or automatic metrics (e.g., BLEU), this method offers a cost-effective and scalable alternative, achieving strong alignment with human evaluations when using advanced models like GPT-4 (Liu et al., 2023). Consequently, LLM-as-a-Judge has seen growing adoption in LLM evaluation tasks (Zheng et al., 2023; Wu et al., 2024; Dubois et al., 2025).

Concerns regarding the high cost, limited transparency, and lack of controllability of proprietary large models (e.g., GPT-4) have driven the adoption of proxy judge models based on open-source LLMs (Wang et al., 2023; Kim et al., 2023, 2024; Li et al., 2024). To equip smaller open-source models with evaluation capabilities comparable to those of advanced large models, this approach typically employs a high-performing teacher model—such as GPT-4—to generate judge-specific training data for evaluation tasks. The open-source model is then fine-tuned on this data to create a specialized proxy judge model. By constructing high-quality judge datasets and applying techniques such as weight merging (Kim et al., 2024), the resulting proxy judge models can achieve performance close to that of large models like GPT-4 (Kim et al., 2023; Li et al., 2024), enabling broad practical adoption.

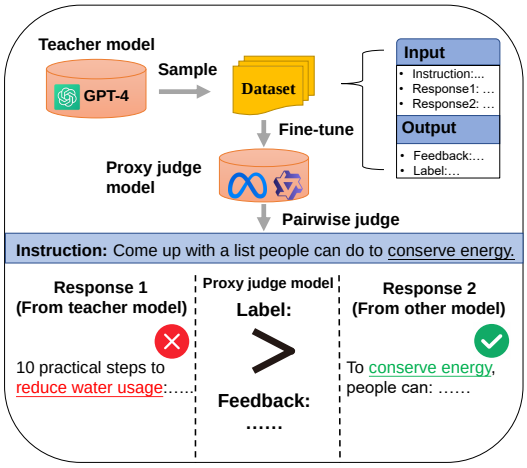


Figure 1: Illustration of the teacher preference bias. The evaluation task is a pairwise ranking, where the proxy judge model generates evaluation feedback and a label indicating which of the two responses is better.

Despite the advantages of proxy judge models, we identify a key limitation: these models exhibit a significant bias favoring responses generated by the teacher model, regardless of their actual quality. We term this phenomenon teacher preference bias (*cf.* Figure 1). This bias stems from the self-preference bias of the teacher model itself (Ye et al.,

†Corresponding author.
[1]Code is available at https://github.com/Liuz233/AGDe-Judge.

2024a; Chen et al., 2025), which is contained in the training data and subsequently captured by the proxy judge model through fine-tuning. To the best of our knowledge, teacher preference bias has not been previously identified or systematically investigated. Our experiments confirm the presence of this bias, which arises from both the judge labels and feedback included in the training data. As GPT-4 is a dominant and highly capable model, many existing methods rely exclusively on data generated by GPT-4 (Li et al., 2024; Kim et al., 2023, 2024), further amplifying the spread of this bias and posing a substantial challenge to the fairness and accuracy of proxy judge models.

To mitigate teacher preference bias, we propose to introduce an additional, smaller LLM, termed as assistant model, into the training pipeline of proxy judge models. These models do not have a biased preference toward the teacher model's responses, and are relatively inexpensive with decent evaluation capability. Within this framework, we leverage the assistant model to debias the teacher model-generated data from two sources: biased labels and biased feedback. Firstly, we aim to identify and filter out the instances with biased labels leveraging information from the reliable labels agreed by the assistant model. Then, we aim to debias the feedback by reducing the teacher model's tendency to overemphasize minor or superficial issues.

To this end, we propose a three-stage framework, AGDe-Judge (Assistant-Guided Debiasing for Judge Models), to mitigate teacher preference bias in proxy judge models. First, we filter out biased labels using an implicit reward margin (Rafailov et al., 2024), which is derived from consensus labels between the teacher and assistant models. Second, we leverage the assistant model to identify severe flaws in responses, thereby counteracting the overemphasis on minor or superficial issues in the feedback. Finally, we fine-tune the student model using the refined data to obtain the final proxy judge model. Experimental results show that AGDe-Judge effectively reduces teacher preference bias while maintaining high evaluation accuracy across six standard benchmarks. In summary, the main contributions of our paper are as follows:

- We identify and define a previously unstudied critical limitation in existing proxy judge models, the teacher preference bias.
- We introduce an assistant LLM to tackle this bias and propose a three-stage approach that tackles biased labels and biased feedback.
- We conduct extensive experiments validating the strong effectiveness of our approach both in debiasing and evaluation performance.

2 Related Work

LLM-as-a-Judge Zheng et al. (2023) proposed LLM-as-a-Judge, leverages powerful LLMs (e.g., GPT-4) to evaluate responses to open-ended questions, offering a scalable alternative to costly human annotators and limited traditional methods. Studies show LLMs achieve high agreement with human experts (Ashktorab et al., 2024; Bavaresco et al., 2024), driving its adoption across various tasks (Zhu et al., 2023; Cui et al., 2023; Bai et al., 2023). However, this paradigm suffers from biases, including position bias (Shi et al., 2025), verbosity bias (Chen et al., 2024), and self-preference bias, where LLMs favor their own responses (Li et al.; Panickssery et al., 2024). These biases undermine the reliability and fairness of LLM-as-a-Judge evaluations (Ye et al., 2024a).

Proxy Judge Models Due to the high cost, limited transparency, and lack of controllability associated with proprietary large models, fine-tuning open-source models to serve as judge models—commonly referred to as proxy judge models—has become an increasingly popular alternative (Wang et al., 2023; Zhu et al., 2023; Kim et al., 2023, 2024). This approach typically involves leveraging a powerful proprietary model (mostly GPT-4), known as the teacher model, to construct evaluation-specific training data. However, the self-preference bias implicitly embedded in the teacher model's outputs may be inherited by the proxy judge model, introducing subtler forms of bias. Despite its significance, this issue has not yet been systematically studied.

Debiasing Approaches Various debiasing approaches have been proposed to address the widespread biases in LLM-as-a-Judge. Discussion-based methods mitigate the bias of a single LLM by leveraging multiple LLMs to engage in debate and deliberation (Khan et al., 2024; Li et al.). Adversarial methods introduce structured rationale pairs (Ye et al., 2024b), or construct specially curated debiasing datasets (Park et al., 2024). However, addressing the previously unstudied teacher preference bias calls for new debiasing methods specifically tailored to this issue.

3 Preliminary Study

Problem Definition LLM-as-a-judge is an efficient and effective strategy for evaluating LLM-generated responses to compare LLM performance. We focus on the most common and effective evaluation setting of *pairwise ranking* (Kim et al., 2024), where the judge LLM evaluates and rank pairs of LLM responses. Let M denote the judge LLM. Given an input tuple consisting of a instruction q, two responses r_0 and r_1 generated by LLMs to be evaluated, and an auxiliary input e which includes evaluation criteria and reference answers (Kim et al., 2024), the judge LLM generates a textual feedback v to illustrate the reason behind its judgment, and a preference label $y \in \{0, 1\}$ to indicate which response is better. Formally,

$$(v, y) = M(q, r_0, r_1, e). \tag{1}$$

Due to the limitations of proprietary large judge models (e.g., GPT-4), such as high cost, limited transparency and controllability, researchers have increasingly turned to proxy judge models as a practical alternative. A common approach begins by constructing a training dataset using outputs from the proprietary model, referred to as the teacher model M_t, denoted as $\mathcal{D}_t = \{(q, r_0, r_1, e), (v_t, y_t)\}$, where v_t and y_t are the textual feedback and label generated by M_t. An open-source model is then fine-tuned on $\mathcal{D}_t$ to obtain a proxy judge model, referred to as the student model M_s, which is used for evaluation tasks. This method of constructing proxy judge models has demonstrated strong performance comparable to that of the teacher model (Kim et al., 2024). Formally, denoting the feedback and label generated from proxy judge model as v_s and y_s,

$$(v_s, y_s) = M_s(q, r_0, r_1, e). \tag{2}$$

Teacher Preference Bias Despite the advantages of proxy judge models, we identify a critical issue in pairwise ranking: when *one response is generated by the teacher model* and the other by an unrelated model, the proxy judge often exhibits a *biased preference for the teacher model's response*, which is much larger than the actual probability of the teacher model's response being better . We refer to this phenomenon as teacher preference bias. This bias stems from the self-preference bias (Ye et al., 2024a; Chen et al., 2025) captured in $\mathcal{D}_t$ generated by the teacher model who often favors its own responses. Denoting the ground-truth label as $y_{gt} \in \{0, 1\}$, and the response index generated by M_t as $y_{tg} \in \{0, 1\}$, this bias can exhibit as

$$P(y_s = y_{tg}) \gg P(y_{gt} = y_{tg}). \tag{3}$$

This issue poses a significant challenge to building reliable proxy judge models. First, existing evaluation datasets (e.g., Preference Collection (Kim et al., 2024), AutoJ-pairwise (Li et al., 2024), JudgeLM (Zhu et al., 2023)) heavily rely on the state-of-the-art teacher model GPT-4, which is widely used due to its strong alignment with human preferences (Liu et al., 2023). Moreover, the response pairs in these datasets often include outputs generated by GPT-4, which creates the condition for introducing such bias. Also, many evaluation benchmarks involve direct comparisons between a GPT-4 response and a response from another model (e.g., MT-Bench (Zheng et al., 2023), Chatbot-Arena (Chiang et al., 2024), Reward-Bench (Lambert et al., 2024)). As a result, the proxy judge model can easily learn and exhibit this latent bias. In the following section, we systematically demonstrate the existence of this bias.

Evaluation Setup To demonstrate the existence of teacher preference bias, we fine-tune proxy judge models using GPT-4 as the teacher model and evaluate their biased preference toward GPT-4-generated responses. The **bias evaluation** is primarily conducted on the following datasets:

- **OffsetBias** (filtered) (Park et al., 2024): A representative evaluation dataset on the bias for LLM-as-a-Judge. The worse responses are generated by GPT-4, crafted intentionally through adversarial prompting techniques such as providing misleading or off-topic instructions, or explicitly encouraging the inclusion of factual errors or incomplete answers. In contrast, the better responses come from other models. Lower accuracy in this dataset show more severe bias.
- **MT-Bench Human Judge** (filtered) (Zheng et al., 2023): Uses the 80 prompts from MT-Bench along with 3.3k pairs of model responses annotated with human preferences. We extract a subset of MT-Bench Human Judge consisting of instances where one response is from GPT-4 and the other from a different model. To evaluate the bias, we measure the extent to which the proxy judge favors GPT-4 responses compared to the ground-truth human labels.

Dataset Filtering	Proxy Judge Model	AutoJ-pairwise			Preference Collection		
		MT-Bench	Arena-Human	OffsetBias	MT-Bench	Arena-Human	OffsetBias
No Filtering	GPT3.5-Mistral-7B	0.695	0.565	**0.269**	0.681	0.538	0.314
		(+0.7%)	(+3.2%)	(-32.3%)	(+7.8%)	(+8.7%)	(-19.4%)
	GPT4-Mistral-7B	0.700	0.583	0.182	**0.734**	**0.585**	0.253
Same-Label Filtering	GPT3.5-Mistral-7B	0.712	0.563	0.230	0.696	0.533	**0.320**
		(+2.2%)	(+4.8%)	(-21.7%)	(+4.2%)	(+5.4%)	(-22.8%)
	GPT4-Mistral-7B	**0.728**	**0.590**	0.180	0.725	0.562	0.247

Table 1: Evaluation accuracy of Mistral-7B-Instruct-v0.3 fine-tuned on the AutoJ-pairwise and Preference Collection datasets under different filtering condition. The best accuracy for each benchmark is **bolded**.

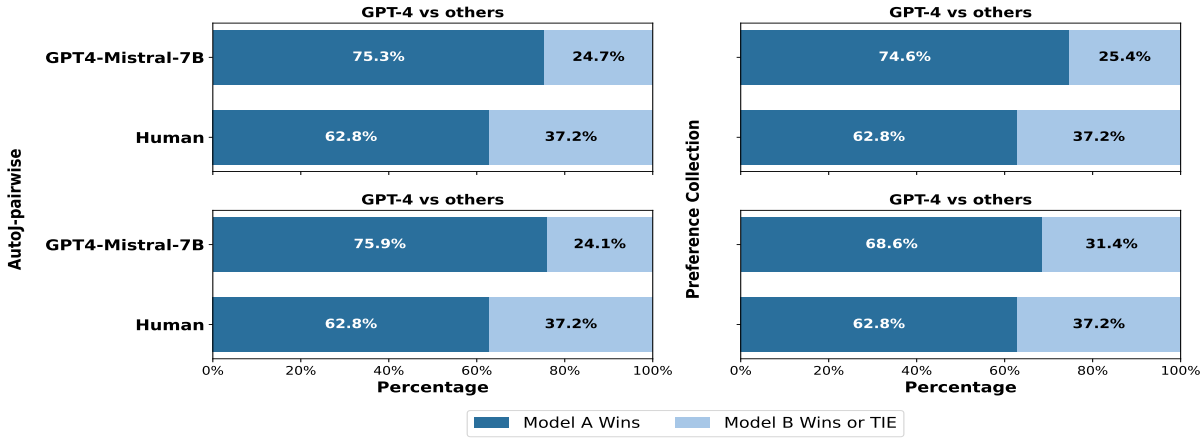


Figure 2: Results of GPT4-Mistral-7B and Human judgments on MT-Bench Human Judge (filtered). The models were trained on AutoJ-pairwise (left) and Preference Collection data (right). The upper row uses all training samples; the lower row uses only those where GPT-4 and GPT-3.5-Turbo assign the same preference label.

Additionally, we employ two **standard benchmarks**, the full **MT-Bench Human Judge** and **Arena-Human** (Chiang et al., 2024), from which we randomly sample evaluation instances to examine the quality of the proxy judge model.

For **fine-tuning datasets**, we employ **AutoJ-pairwise** and **Preference Collection**. Both are constructed under GPT-4 with evaluated responses from GPT-4. In addition to using GPT-4 as the teacher model, we investigate the severity of teacher preference bias by introducing **an alternative teacher model: GPT-3.5-Turbo**. We prompt GPT-3.5-Turbo with the same inputs used in the fine-tuning datasets to generate feedback and labels. Comparing the performance of proxy judge models fine-tuned with GPT-3.5-Turbo versus GPT-4 as teacher enables us to evaluate the extent of bias toward GPT-4-generated responses. For **base proxy judge model**, we use the open-source **Mistral-7B-Instruct-v0.3** (Jiang et al., 2023).

Furthermore, we investigate **different sources of bias**. We hypothesize that bias may arise from both the preference labels and the textual feedback. To isolate the effect of feedback, we introduce an additional setting where we retain only training instances for which GPT-4 and GPT-3.5-Turbo teacher models assign identical preference labels. This allows us to assess the extent of bias attributable specifically to the feedback. More details regarding dataset processing, fine-tuning, and generating procedures are in Appendix A.

Evaluation Results Table 1 shows the bias and accuracy results for proxy judge models with different teacher models. On standard benchmarks, the model trained with GPT-4 as the teacher (GPT4-Mistral-7B) achieves the highest accuracy, outperforming GPT3.5-Mistral-7B. This highlights GPT-4's strong alignment with human evaluations and supports its use as a teacher model for generating fine-tuning data. However, on the bias evaluation set OffsetBias, GPT4-Mistral-7B shows a notable performance drop compared to GPT3.5-Mistral-7B. This suggests that GPT4-Mistral-7B incorrectly favors GPT-4 responses even when they are worse. Together, these findings indicate that, despite its overall effectiveness, the proxy judge model tends to over-rank responses from its teacher model—a phenomenon we refer to as *teacher preference bias*.

Results using identical labels from both teacher models are also presented below Table 1, showing a trend consistent with the unfiltered results: while GPT4-Mistral-7B achieves higher accuracy on standard benchmarks, it still demonstrates a clear biased preference toward responses from GPT-4, con-

firming the presence of teacher preference bias. Notably, the bias performance gap in the feedback-only setting is smaller than that in the joint setting. Taken together, these findings indicate that both labels and feedback in the training data contribute to the teacher preference bias.

Figure 2 presents the results on MT-Bench Human Judge (filtered), offering an complementary metric, namely by comparing the proportion of GPT-4 responses labeled as winner by judge models with the actual proportion labeled as winner by human annotators, thereby quantifying the degree of teacher preference bias. We can clearly observe that, under different training datasets and sampling settings, the win ratio assigned to GPT-4 by GPT4-Mistral-7B is substantially higher than that assigned by human annotators. This provides another piece of intuitive evidence that reinforces the existence of teacher preference bias.

Case Study To further investigate the causes of teacher preference bias, we conduct a case study. We first observe the patterns of GPT-4's self-preference bias in the training data, as illustrated in Table 12. The case reveals that GPT-4's feedback is not solely grounded in core evaluation criteria such as relevance, completeness, and factual accuracy. Instead, it also emphasizes features commonly associated with GPT-4's own outputs—such as informativeness, extensive citations, and greater depth of reasoning. Next, we examine the failure patterns of proxy judge models with respect to teacher preference bias, as shown in Table 13. We find that the model tends to prioritize stylistic and content-rich features like vividness and depth of knowledge—features typical of GPT-4 outputs—while overlooking critical flaws such as topical irrelevance. This case shows that self-preference bias present in the training data is captured by the proxy judge model during fine-tuning.

4 Methodology

After identifying the teacher preference bias, our goal is to mitigate this bias in proxy judge models while maintaining their strong evaluation performance. We incorporate an additional smaller LLM, referred to as the assistant model, into the fine-tuning process. The assistant model is cost-effective, exhibits reliable evaluation capabilities, and remains unaffected by the teacher model's bias. Building on our earlier observations, the proposed framework aims to reduce teacher preference bias by targeting two key sources sequentially: first, by filtering biased labels, and then by addressing biased feedback. We propose AGDe-Judge, a three-stage debiasing framework to achieve this goal.

Stage 1: Label Filtering by Reward Margin We observe that labels in the dataset $\mathcal{D}_t$ can be incorrect and biased. Therefore, the first stage of our framework focuses on filtering out incorrectly labeled instances. Inspired by Deng et al. (2025), we assign a reward score to each candidate response in an evaluation instance and retain only those instances with a sufficiently large reward margin between the better and worse responses—indicating a clear quality difference. To ensure the reward is reliable and corrects potential bias from the teacher model, we leverage instances where the teacher and assistant models agree on the ranking, rather than relying solely on the teacher model's judgment.

Specifically, we first construct an additional training dataset by prompting the assistant model with the same instructions and candidate responses, generating new labels and feedback. We then select samples where both the teacher and assistant models provide consistent label annotations, denoted as $\mathcal{D}_a$. These samples are used to train an auxiliary DPO model M_i, using the following loss function:

$$\mathcal{L}_i = -\mathbb{E}_{\mathcal{D}_a}\Bigg[\log\sigma\Bigg(\beta\frac{M_i(r_y|q)}{M_{\text{ref}}(r_y|q)} - \beta\frac{M_i(r_{1-y}|q)}{M_{\text{ref}}(r_{1-y}|q)}\Bigg)\Bigg], \quad (4)$$

where M_{ref} is the reference assistant model. Next, we compute the implicit reward margin between the better and worse responses, r_y and r_{1-y}:

$$t = \log\frac{M_i(r_y|q)}{M_{\text{ref}}(r_y|q)} - \log\frac{M_i(r_{1-y}|q)}{M_{\text{ref}}(r_{1-y}|q)}. \quad (5)$$

By applying a threshold T on this reward margin, we filter out instances with potentially biased labels in both datasets. The remaining better-quality samples are then passed to the next stage.

Stage 2: Feedback Debiasing by Assistant-Critique Aggregation For instances with filtered labels, the feedback provided by the teacher model may still exhibit significant bias—overemphasizing minor features characteristic of its own responses while overlooking more critical aspects of response quality (see Section 3). In contrast, the assistant model's feedback is less prone to such self-preference bias.

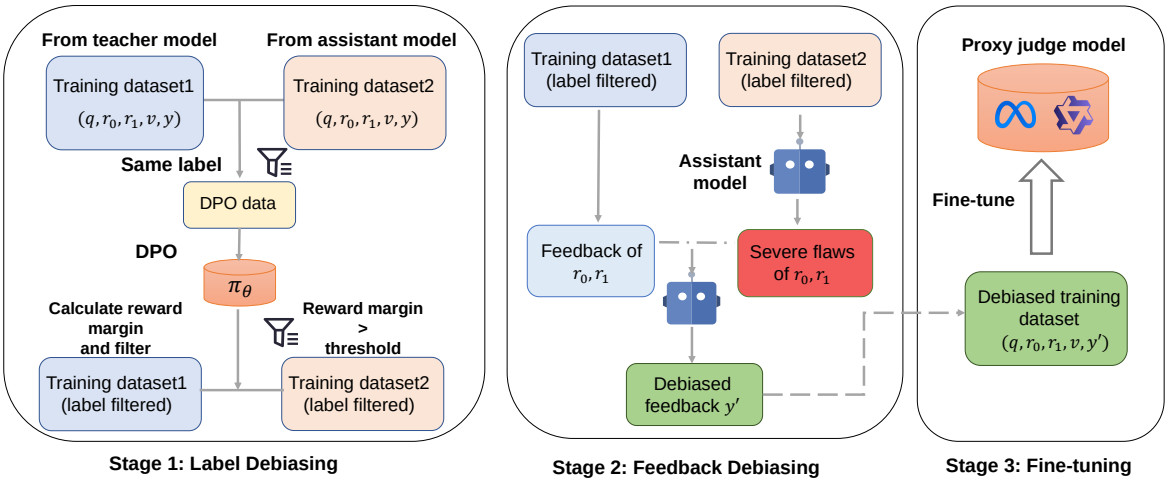


Figure 3: Illustration of three-stage debiasing framework AGDe-Judge. q, r_0, r_1, v, y denote the inputs and outputs of the pairwise ranking task, following the same notation as in Section 3.

To address this, we utilize the assistant model to identify severe flaws in both candidate responses. These identified flaws serve as anchors to guide the assistant model in mitigating the biased focus of the original feedback on minor or superficial features. The assistant model is then prompted to generate revised feedback by integrating the detected flaws with the original teacher feedback, resulting in a more balanced and critical evaluation. Formally,

$$c_a = M_a(q, r_0, r_1, w), \tag{6}$$
$$v_a = M_a(q, r_0, r_1, v, c_a, e, o), \tag{7}$$

where M_a is assistant model, c_a is the critique under prompt w, and v_a is the revised feedback under original feedback v, c_a, and prompt o.

Stage 3: Fine-tuning After the first two stages, we obtain de-biased labels and feedback. We then fine-tune the proxy judge model M_s using the following loss function. Denoting the dataset obtained from the first two stages as $\mathcal{D}_j = \{q, r_0, r_1, e, v_a, y_a\}$:

$$\mathcal{L}_j = -\mathbb{E}_{\mathcal{D}_j}\left[\log \frac{M_s(v_a, y_a|q, r_0, r_1, e)}{M_{\text{ref}}(v_a, y_a|q, r_0, r_1, e)}\right]. \tag{8}$$

5 Experiments

Datasets and Models To evaluate the extent of teacher preference bias across different methods, we use two **bias evaluation datasets**: **OffsetBias** (filtered) and **MT-Bench Human Judge**(filtered), processed following the same protocol as in Section 3. In addition, we assess the accuracy of proxy judge models using **six standard benchmarks**: **Preference-Bench** (Kim et al., 2024), **MT-Bench Human Judge**, **Reward-Bench** (Lambert et al., 2024), **Arena-Human**, **UltraFeedback Binarized** (Cui et al., 2023), **JudgeLM** (Zhu et al., 2023). See Appendix A.1 for more details.

For **fine-tuning datasets**, we employ **AutoJ-pairwise**. We use **GPT-4** as the **teacher model**. To validate the effectiveness of our proposed framework AGDe-Judge, we experiment with two different **assistant models**: **GPT-3.5-Turbo** and **Qwen2.5-7B-Instruct** (Qwen, 2024). In the DPO phase of Stage 1, we adopt **Llama-2-7b-hf** (Touvron et al., 2023) as the base model. For the **proxy judge model**, we use the open-source **Mistral-7B-Instruct-v0.3** as the backbone.

Compared Methods In addition to directly using the base model and training the proxy judge model with conventional methods, we also compare our approach with data-centric and model-centric optimization baselines.

- **Naive Mix**, directly mix the training data generated by the teacher model and the assistant model without any further processing.
- **Teacher-Only Margin Filter**, after computing the implicit reward margin in Stage 1, we use the filtered training data from the teacher model as the final training set.
- **Weight Merging** (Rame et al., 2023), involves two proxy judge models, θ_t and θ_a, which are fine-tuned separately on the training data generated by the teacher model and the assistant model, respectively. Then, we obtain the final proxy judge model via linear merging: $\theta_{final} = \alpha \cdot \theta_t + (1 - \alpha) \cdot \theta_p$, where we experiment by using $\alpha = 0.5$.

	Preference-Bench	MT-Bench	Reward-Bench	Arena-Human	UltraFeedback	JudgeLM	OffsetBias
Mistral-7B	0.669	0.571	0.835	0.462	0.554	0.616	0.322
GPT3.5-Mistral-7B	**0.874**	0.695	0.931	0.565	0.681	0.735	0.269
GPT4-Mistral-7B	0.721	0.700	0.962	0.583	0.735	0.760	0.182
Naive Mix	0.788	0.665	0.953	0.598	0.687	0.733	0.194
Teacher-Only Margin Filter	0.831	0.700	**0.964**	0.586	**0.754**	0.783	0.207
Weight Merging	0.853	0.710	0.941	0.594	0.707	0.761	0.293
AGDe-Judge	0.832	**0.715**	0.947	**0.598**	0.744	**0.792**	**0.391**
	Preference-Bench	**MT-Bench**	**Reward-Bench**	**Arena-Human**	**UltraFeedback**	**JudgeLM**	**OffsetBias**
Mistral-7B	0.669	0.571	0.835	0.462	0.554	0.616	0.322
Qwen2.5-Mistral-7B	0.852	0.715	0.915	0.565	0.702	0.753	0.360
GPT4-Mistral-7B	0.721	0.700	**0.962**	0.583	0.735	0.760	0.182
Naive Mix	0.840	0.716	0.926	0.536	0.713	0.752	0.335
Teacher-Only Margin Filter	0.812	0.695	0.961	0.598	0.747	0.760	0.204
Weight Merging	**0.856**	**0.720**	0.942	0.582	0.742	0.766	0.340
AGDe-Judge	0.833	0.716	0.945	**0.608**	**0.748**	**0.777**	**0.373**

Table 2: Evaluation results on standard benchmarks and the teacher preference bias test dataset. The upper part is when the assistant model is **GPT-3.5-Turbo**, and that for the lower part is **Qwen2.5-7B-Instruct**. **Bolded** and underlined numbers denote the best and the second-best value.

	Label Debias Only	Feedback Debias Only	AGDe-Judge
Preference-Bench	0.822	0.798	**0.832**
MT-Bench	0.702	0.679	**0.715**
Reward-Bench	0.946	0.930	**0.947**
Arena-Human	0.593	0.592	**0.598**
UltraFeedback	0.712	0.725	**0.744**
JudgeLM	0.758	0.774	**0.792**
OffsetBias	0.274	0.335	**0.391**

Table 3: Results of ablation study. Each row corresponds to one benchmark and the best accuracy for each benchmark is **bolded**.

	Naive Concatenation	Rephrasing	AGDe-Judge
Preference-Bench	**0.868**	0.840	0.832
MT-Bench	0.692	0.696	**0.715**
Reward-Bench	0.932	0.945	**0.947**
Arena-Human	0.570	0.580	**0.598**
UltraFeedback	0.690	**0.747**	0.744
JudgeLM	0.762	0.783	**0.792**
OffsetBias	0.271	0.320	**0.391**

Table 4: Results of feedback debiasing with different prompt strategies.

There are also some debiasing methods, such as multi-agent debate methods (Khan et al., 2024; Li et al.), constructing debiasing datasets (Park et al., 2024) and Con-J (Ye et al., 2024b), that were not included as baselines for comparison. This is because these methods are not directly applicable to our problem setting, which involves only teacher-generated training data and an assistant model, and thus including them as baselines would not constitute a fair comparison. The detailed reasons and analysis are provided in Appendix A.3 and A.4.

Except for **MT-Bench Human Judge** (filtered), where results are reported as win/loss ratios, all other datasets use **accuracy** as the evaluation metric. For implementation details of fine-tuning and DPO, please refer to Appendix A.

5.1 Results

Table 2 present the results when using GPT-3.5-Turbo and Qwen2.5-7B-Instruction as the assistant models, respectively. We can observe the following key findings:(1) AGDe-Judge outperforms all baselines, achieving the best results on most benchmarks, and simultaneously achieves the highest accuracy on OffsetBias. This indicates its strong ability to mitigate teacher preference bias, which is attributed to its effective removal of bias from both labels and feedback, thereby prevents the hidden self-preference bias in the training data from influencing the proxy judge model. Figure 4 further demonstrates the effectiveness of AGDe-Judge in mitigating teacher preference bias. (2) Naive Mix performs poorly because it retains biased and conflicting labels and feedback, harming model performance. In contrast, AGDe-Judge ensures data quality by filtering out bias and inconsistencies. (3) Teacher-Only Margin Filter improves over basic teacher-trained models by removing biased labels, but biased feedback still causes teacher preference bias. AGDe-Judge further mitigates this by using the assistant model to highlight key flaws, leading to better results. (4) Weight-Merging, a model-centric approach, combines teacher and assistant models at the parameter level and improves accuracy, but fails to fully remove bias. AGDe-Judge adopts a data-centric strategy, directly eliminating bias from training data and achieving superior debiasing and evaluation performance. (5) The results in Figure 6 indicate that introducing the assistant model in AGDe-Judge does not introduce additional teacher preference bias.

5.2 In-depth Analysis

Ablation Studies To validate the effectiveness of each component within our framework, we con-

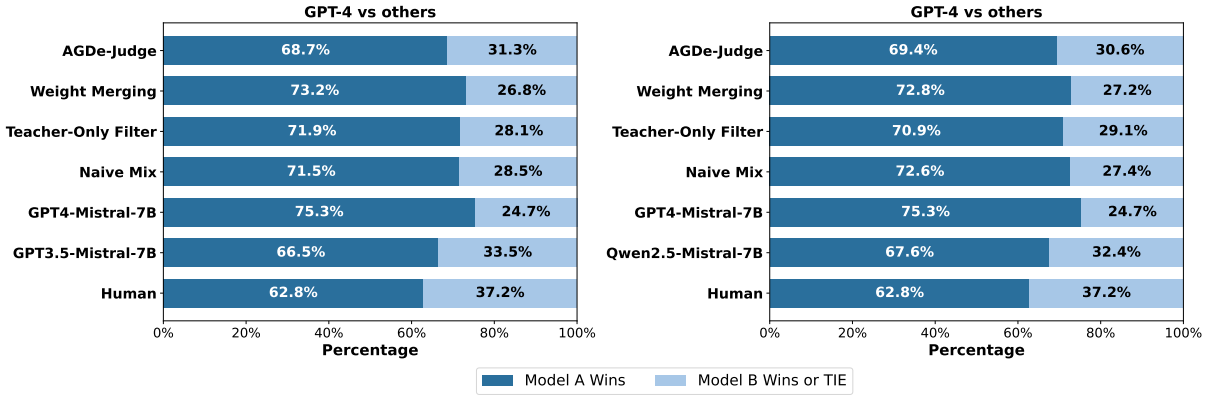


Figure 4: Result on MT-Bench Human Judge (filtered). All methods in the left subfigure use GPT-3.5-Turbo as the assistant model, while those in the right subfigure use Qwen2.5-7B-Instruction.

	Threshold=0	Threshold =5	Threshold=10
Preference-Bench	0.799	0.832	**0.850**
MT-Bench	0.708	**0.715**	0.712
Reward-Bench	0.924	**0.947**	0.937
Arena-Human	0.572	**0.598**	0.586
UltraFeedback	0.707	**0.744**	0.717
JudgeLM	0.753	**0.792**	0.736
OffsetBias	0.277	**0.391**	0.239

Table 5: Results of different implicit reward margin threshold. The best accuracy for each benchmark is **bolded**, and the second-best value is underlined.

duct the following ablation studies: (1) **Label Debiasing Only**: we remove the Stage 2 feedback debiasing process and retain only the label filtering step from Stage 1; (2) **Feedback Debiasing Only**: we skip the label debiasing in Stage 1 and directly apply feedback refinement to the original training datasets. We use GPT-3.5-Turbo as assistant model.

From the results shown in Table 3, we observe that compared to the full AGDe-Judge framework, performing only label debiasing or only feedback debiasing leads to significant performance degradation, demonstrating the necessity of implementing debiasing measures for both labels and feedback. Label debiasing alone results in a greater decline on the OffsetBias test set, while feedback debiasing alone causes a more substantial accuracy drop on standard benchmarks. This indicates that in the AGDe-Judge framework, feedback debiasing plays a more critical role in reducing bias, whereas label debiasing is more essential for improving accuracy.

Comparison of Different Prompting Strategy Design To evaluate the impact of different prompting strategies on feedback debiasing in Stage 2, we experiment with several prompt designs: (1) **Naive Concatenation**: directly present the feedback from both the teacher and assistant models to the assistant model, prompting it to generate a final evaluation; (2) **Rephrasing**: directly prompt the assistant model to rephrase the feedback originally provided by the teacher model; (3) **AGDe-Judge**: prompt the assistant model to explicitly identify severe flaws in the responses, and then generate a final evaluation that integrates the original feedback from the teacher model, which is adopted by AGDe-Judge. For detail prompt, refer to Appendix C.

Table 4 presents the results, showing that AGDe-Judge outperforms other prompt strategies. This is because AGDe-Judge's Stage 2 leverages the assistant model to identify significant and severe flaws in the candidate responses, it can compensates for the teacher model's tendency to overemphasize minor and superficial issues. As a result, the final feedback is more focused on critical aspects, more comprehensive, and free from bias, leading to higher overall quality.

Sensitivity to the Implicit Reward Margin In this part, to assess the impact of different implicit reward margin thresholds used in Stage 1 on label filtering, we conduct experiments using three threshold values: 0, 5, and 10. Figure 7 shows the distribution of implicit reward margins in labels generated by GPT-4 and GPT-3.5-Turbo. From the results in Table 5, we can infer that when labels are filtered using a threshold of 5, the resulting proxy judge model achieves the best performance. This is attributed to the following reasons: (1) When the threshold is lower, the debiasing effect on labels is weak, leading to relatively poor quality of the remaining filtered data. (2) When the threshold is set too high, the number of remaining samples after filtering decreases sharply, making it insufficient to effectively train the proxy judge model.

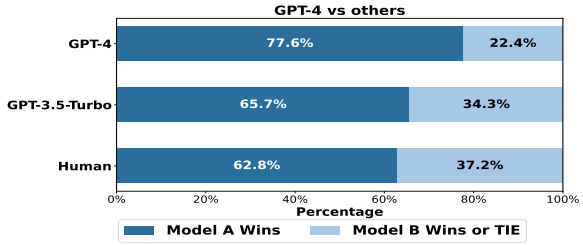


Figure 5: Annotation results from GPT-4, GPT-3.5-Turbo API, and human on MT-Bench Human Judge (filtered) for GPT-4 vs others.

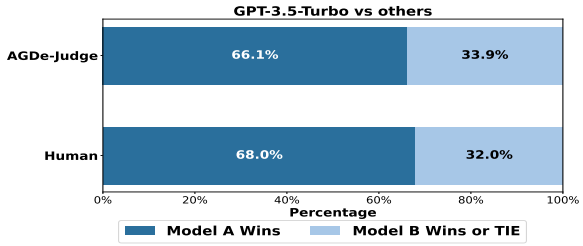


Figure 6: Result on MT-Bench Human Judge (filtered) of GPT-3.5-Turbo vs others.

Potential Biases Introduced by the Assistant Model To investigate whether the introduction of the assistant model amplifies the existing teacher preference bias or introduces additional biases, we conducted further experiments on the filtered MT-Bench dataset. Figure 5 illustrates the degree of bias of GPT-3.5-Turbo toward GPT-4's responses. By comparing the win rate of GPT-4's responses labeled by GPT-3.5-Turbo against those labeled by human annotators, we observe that the selected assistant model (GPT-3.5-Turbo) exhibits minimal bias toward the teacher, even though they belong to the same model family. This result can be attributed to the fact that the assistant model we selected is generally smaller in scale, making it less likely to capture the teacher model's stylistic patterns and preferences—thereby reducing the risk of bias toward the teacher.

In addition, Figure 6 shows that AGDe-Judge exhibits no significant preference for the assistant model. Therefore, we conclude that the introduction of assistant models such as GPT-3.5-Turbo neither amplifies the bias toward the teacher model nor introduces potential new biases, thus supporting the robustness of our evaluation framework.

6 Conclusion

This work investigates a previously overlooked bias: teacher preference bias. Through extensive experiments and analysis, we demonstrate that training datasets generated by teacher models inherently contain self-preference bias in both labels and feedback, which in turn induces teacher preference bias in proxy judge models. To mitigate this, we introduce AGDe-Judge, a three-stage framework that filters biased labels using an implicit reward margin and refines biased feedback with the help of an assistant model. Our experimental results show the superior performance of our approach over several existing baselines across multiple standard and bias evaluation benchmarks.

Limitations

Conducting Experiments with Additional Teacher Models All our experiments used GPT-4 as the teacher model, as it is the dominant and highly capable model most widely used for generating training data. However, the extent of teacher preference bias when using other advanced large language models, such as the Claude series or DeepSeek, as teacher models remains to be further explored.

Proposing Solutions from Alternative Perspectives Our AGDe-Judge framework primarily focuses on debiasing training data. Future work could explore alternative optimization strategies from the perspectives of model architecture or training methodologies to further mitigate teacher preference bias.

Deeper Investigation into the Sources of Bias Our work demonstrates that the teacher preference bias in proxy judge models primarily stems from self-preference bias implicitly embedded in the labels and feedback of the training data. However, the underlying causes of self-preference bias have not yet been thoroughly investigated or substantiated. Studying the root causes of bias could facilitate more effective solutions for addressing both self-preference bias and teacher preference bias.

Ethical Consideration

The case study shown in the Appendix D includes responses from LLMs, some of which may contain non-factual or harmful information.

Acknowledgement

This work is supported by the National Natural Science Foundation of China (62272437).

References

Zahra Ashktorab, Michael Desmond, Qian Pan, James M Johnson, Martin Santillan Cooper, Elizabeth M Daly, Rahul Nair, Tejaswini Pedapati, Swapnaja Achintalwar, and Werner Geyer. 2024. Aligning human and llm judgments: Insights from evalassist on task-specific evaluations and ai-assisted assessment strategy preferences. *arXiv preprint arXiv:2410.00873*.

Yushi Bai, Jiahao Ying, Yixin Cao, Xin Lv, Yuze He, Xiaozhi Wang, Jifan Yu, Kaisheng Zeng, Yijia Xiao, Haozhe Lyu, and 1 others. 2023. Benchmarking foundation models with language-model-as-an-examiner. *Advances in Neural Information Processing Systems*, 36:78142–78167.

Anna Bavaresco, Raffaella Bernardi, Leonardo Bertolazzi, Desmond Elliott, Raquel Fernández, Albert Gatt, Esam Ghaleb, Mario Giulianelli, Michael Hanna, Alexander Koller, and 1 others. 2024. Llms instead of human judges? a large scale empirical study across 20 nlp evaluation tasks. *CoRR*.

Guiming Chen, Shunian Chen, Ziche Liu, Feng Jiang, and Benyou Wang. 2024. Humans or llms as the judge? a study on judgement bias. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8301–8327.

Wei-Lin Chen, Zhepei Wei, Xinyu Zhu, Shi Feng, and Yu Meng. 2025. Do llm evaluators prefer themselves for a reason? *arXiv preprint arXiv:2504.03846*.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E. Gonzalez, and Ion Stoica. 2024. Chatbot arena: An open platform for evaluating llms by human preference. *Preprint*, arXiv:2403.04132.

Ganqu Cui, Lifan Yuan, Ning Ding, Guanming Yao, Wei Zhu, Yuan Ni, Guotong Xie, Zhiyuan Liu, and Maosong Sun. 2023. Ultrafeedback: Boosting language models with high-quality feedback. *Preprint*, arXiv:2310.01377.

Xun Deng, Han Zhong, Rui Ai, Fuli Feng, Zheng Wang, and Xiangnan He. 2025. Less is more: Improving LLM alignment via preference data selection. *CoRR*, abs/2502.14560.

Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. 2025. Length-controlled alpacaeval: A simple way to debias automatic evaluators. *Preprint*, arXiv:2404.04475.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. Mistral 7b. *Preprint*, arXiv:2310.06825.

Akbir Khan, John Hughes, Dan Valentine, Laura Ruis, Kshitij Sachan, Ansh Radhakrishnan, Edward Grefenstette, Samuel R Bowman, Tim Rocktäschel, and Ethan Perez. 2024. Debating with more persuasive llms leads to more truthful answers. In *Proceedings of the 41st International Conference on Machine Learning*, pages 23662–23733.

Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and 1 others. 2023. Prometheus: Inducing fine-grained evaluation capability in language models. In *The Twelfth International Conference on Learning Representations*.

Seungone Kim, Juyoung Suk, and Minjoon Seo. 2024. Prometheus 2: An open source language model specialized in evaluating other language models. In *EMNLP 2024*. Association for Computational Linguistics (ACL).

Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. 2024. Rewardbench: Evaluating reward models for language modeling. *Preprint*, arXiv:2403.13787.

Junlong Li, Shichao Sun, Weizhe Yuan, Run-Ze Fan, Pengfei Liu, and 1 others. 2024. Generative judge for evaluating alignment. In *The Twelfth International Conference on Learning Representations*.

Ruosen Li, Teerth Patel, and Xinya Du. Prd: Peer rank and discussion improve large language model based evaluations. *Transactions on Machine Learning Research*.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-eval: Nlg evaluation using gpt-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522.

OpenAI. 2024. Gpt-4 technical report. *Preprint*, arXiv:2303.08774.

Arjun Panickssery, Samuel Bowman, and Shi Feng. 2024. Llm evaluators recognize and favor their own generations. *Advances in Neural Information Processing Systems*, 37:68772–68802.

Junsoo Park, Seungyeon Jwa, Ren Meiying, Daeyoung Kim, and Sanghyuk Choi. 2024. Offsetbias: Leveraging debiased data for tuning evaluators. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1043–1067.

Qwen. 2024. Qwen2.5: A party of foundation models.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2024. Direct preference optimization: Your language model is secretly a reward model. *NeurIPS*, 36.

Alexandre Rame, Guillaume Couairon, Corentin Dancette, Jean-Baptiste Gaya, Mustafa Shukor, Laure Soulier, and Matthieu Cord. 2023. Rewarded soups: towards pareto-optimal alignment by interpolating weights fine-tuned on diverse rewards. *Advances in Neural Information Processing Systems*, 36:71095–71134.

Lin Shi, Chiyu Ma, Wenhua Liang, Xingjian Diao, Weicheng Ma, and Soroush Vosoughi. 2025. Judging the judges: A systematic study of position bias in llm-as-a-judge. *Preprint*, arXiv:2406.07791.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, and 1 others. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

Yidong Wang, Zhuohao Yu, Wenjin Yao, Zhengran Zeng, Linyi Yang, Cunxiang Wang, Hao Chen, Chaoya Jiang, Rui Xie, Jindong Wang, and 1 others. 2023. Pandalm: An automatic evaluation benchmark for llm instruction tuning optimization. In *The Twelfth International Conference on Learning Representations*.

Tianhao Wu, Weizhe Yuan, Olga Golovneva, Jing Xu, Yuandong Tian, Jiantao Jiao, Jason Weston, and Sainbayar Sukhbaatar. 2024. Meta-rewarding language models: Self-improving alignment with llm-as-a-meta-judge. *arXiv preprint arXiv:2407.19594*.

Jiayi Ye, Yanbo Wang, Yue Huang, Dongping Chen, Qihui Zhang, Nuno Moniz, Tian Gao, Werner Geyer, Chao Huang, Pin-Yu Chen, and 1 others. 2024a. Justice or prejudice? quantifying biases in llm-as-a-judge. In *Neurips Safe Generative AI Workshop 2024*.

Ziyi Ye, Xiangsheng Li, Qiuchi Li, Qingyao Ai, Yujia Zhou, Wei Shen, Dong Yan, and Yiqun Liu. 2024b. Beyond scalar reward model: Learning generative judge from preference data. *arXiv preprint arXiv:2410.03742*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, and 1 others. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems*, 36:46595–46623.

Lianghui Zhu, Xinggang Wang, and Xinlong Wang. 2023. Judgelm: Fine-tuned large language models are scalable judges.

Appendices

A Experimental Details

A.1 Dataset Processing

Preference-Bench (Kim et al., 2024): A benchmark split from the Preference Collection, serving as the in-domain test set for PROMETHEUS models. It consists of 200 prompts, 2,000 response pairs labeled as "win" or "lose", and 200 evaluation criteria.

MT-Bench Human Judge (Zheng et al., 2023): Comprises 80 prompts from MT-Bench and 3.3k pairs of model responses annotated with human preferences. We use the full set after removing tie cases to evaluate accuracy. We filtered samples from the first round of dialogue with labels other than "TIE".

Reward-Bench (Lambert et al., 2024): A benchmark for evaluating alignment with human preferences, covering four domains: chat, chat hard, safety, and reasoning.

Arena-Human (Chiang et al., 2024): Arena-Human is a dataset collected from the Chatbot Arena platform, a crowdsourced platform featuring anonymous battles between chatbots in real-world scenarios (Chiang et al., 2024). Contains 100k pairwise human-labeled comparisons collected from the Chatbot Arena leaderboard, reflecting real-world model preferences. We filtered samples from the first round of dialogue with labels other than "TIE".

UltraFeedback Binarized (Cui et al., 2023): A binarized version of the original UltraFeedback dataset. It includes 64k prompts, with chosen and rejected responses determined by the overall score.

JudgeLM (Zhu et al., 2023): Includes 100k+ high-quality judgment samples generated by GPT-4. We use a subset of test samples that have been manually reviewed and validated by human annotators.

For the all standard benchmarks used to test the accuracy, we randomly selected 1k data for testing and removed the samples labeled as “TIE”.

A.2 Implementation Details

Details for Fine-tuning Mistral-7B-Instruct-v0.3 model For both the AutoJ-pairwise and Preference Collection training datasets, we set the maximum input length to 2048, the learning rate to 1e-5, the number of training epochs to 2, the gradient accumulation steps to 4, and the batch size to 4. We adopt the AdamW optimizer and apply LoRA for efficient fine-tuning. During training, we follow the prompt templates provided with each training dataset. AutoJ-pairwise uses all 3,400 samples with those labeled as "TIE" removed for training. Preference Collection randomly selects 2,000 samples for training. Both datasets are split into training and validation sets with a 9:1 ratio. Detailed prompt formats are described in Appendix C. We conduct all experiments on 8 NVIDIA A100 GPUs, and the same setup is used for the following experiments.

Details for DPO training Llama-2-7b-hf We randomly selected 1,000 samples in Autoj-pairwise with consistent labels for DPO training, and split them into training and test sets with a 9:1 ratio. We set the maximum input length to 1024, the learning rate to 1e-4, the warmup steps to 0.1, the number of training epochs to 3, the gradient accumulation steps to 4, and the batch size to 1.

Details for Evaluation During sampling, we set max tokens to 1024, repetition penalty to 1.03, best_of to 1, temperature to 1.0, and top_p to 0.9. Since smaller open-source models often exhibit limited instruction-following capabilities and occasionally fail to generate well-structured feedback and labels, we follow the sampling strategy proposed in Prometheus 2 (Kim et al., 2024). Specifically, we apply regular expression-based extraction with a maximum of 10 attempts to reliably parse valid outputs.

A.3 Problem Setting

In our problem setting, we only make use of training data generated by a powerful teacher model together with an additional lightweight or low-cost assistant model. This aligns with common practice, as many existing practices rely on pre-existing judge datasets (Kim et al., 2023, 2024; Li et al., 2024) from strong teacher models such as GPT-4, and it also avoids the high computational overhead or high cost of repeatedly invoking such models.

A.4 Baseline Selection Details

Some major debiasing methods are not included as our baselines for comparison, as they are not directly applicable to our problem setting we predefined. In particular, they introduce additional

models or datasets. A detailed explanation is provided below:

Multi-agent debate methods Such methods typically rely on multiple models with different architectures and preferences engaging in debate to mitigate the bias introduced by a single model (Li et al.; Khan et al., 2024). However, they require incorporating several distinct large models of comparable capability, which is not compatible with our problem setting.

constructing adversarial training sets These methods construct responses that exhibit a specific type of bias—for example, answers with superficially higher quality but containing severe errors—as negative samples in a pairwise ranking task, thereby forming adversarial training sets used for fine-tuning the proxy judge model to internalize the ability to detect such errors (Park et al., 2024). However, for addressing teacher preference bias, re-invoking the teacher model to generate negative samples would violate our problem setting; moreover, due to the differences in biases between the assistant and teacher models, it is challenging for the assistant model to effectively generate such negative samples with stylistic characteristics of the teacher model.

Constructing Positive and Negative Judgment Samples Such methods typically rely on manual annotation and repeated sampling to generate correct and incorrect judgment feedback and labels, which are subsequently used for DPO training of the proxy judge model, as in Con-J (Ye et al., 2024b). However, in the absence of human annotation, it is not feasible to deliberately construct incorrect judgment examples that encode teacher preference bias, and therefore this approach cannot be straightforwardly applied to our problem setting.

B Additional Experimental Results

We present some additional experimental results.

B.1 Visualization of Margin Distributions

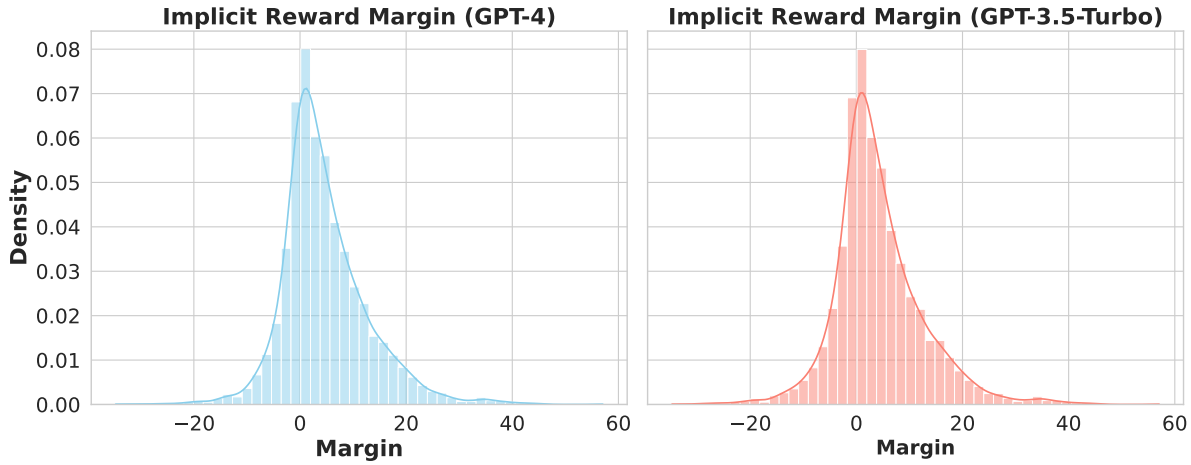


Figure 7: Distribution of implicit reward margins in labels generated by GPT-4 and GPT-3.5-Turbo.

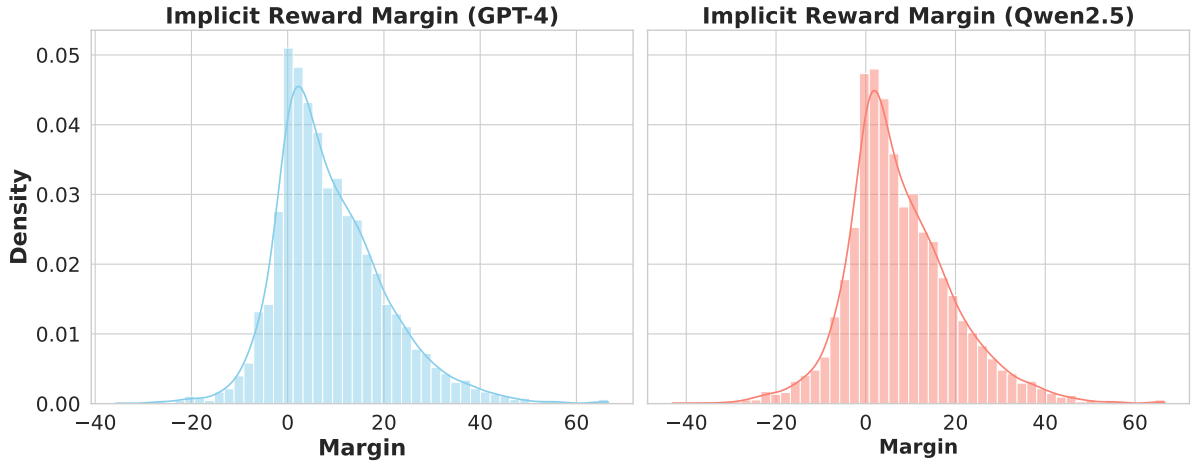


Figure 8: Distribution of implicit reward margins in labels generated by GPT-4 and Qwen2.5-7B-Instruction.

C Prompts

We provided prompts corresponding to different training datasets, as shown in Table 6 and Table 7. Table 8 presents the prompt for the assistant model to identify severe flaws in both responses. Table 9, Table 10, and Table 11 correspond to different feedback optimization strategies, namely: directly generating new feedback based on two evaluations, directly paraphrasing the teacher model's feedback, and generating new feedback based on the teacher model's feedback combined with severe flaws identified by the assistant model.

###Task Description: An instruction (might include an Input inside it), a response to evaluate, a reference answer, and a score rubric representing a evaluation criteria are given. 1. Write a detailed feedback that assess the quality of two responses strictly based on the given score rubric, not evaluating in general. 2. After writing a feedback, choose a better response between Response A and Response B. You should refer to the score rubric. 3. The output format should look as follows: "(write a feedback for criteria) [RESULT] (A or B)" 4. Please do not generate any other opening, closing, and explanations. ###Instruction: {instruction} ###Response A: {response_A} ###Response B: {response_B} ###Reference Answer: {reference_answer} ###Score Rubric: {rubric} ###Feedback:

Table 6: The prompt template used for the proxy judge model trained on the Preference Collection dataset.

You are assessing two submitted responses on a given user's query and judging which response is better. Here is the data:

[BEGIN DATA]
***
[Query]: {instruction}
***
[Response 1]: {response_A}
***
[Response 2]: {response_B}
***
[END DATA]

Here are the instructions to assess and compare the two responses:

1. Pinpoint the key factors to distinguish these two responses.
2. Conclude your comparison by providing a final decision on which response is better. Begin your final decision statement with "So, the final decision is Response 1 / Response 2". Ensure that your decision aligns coherently with the comprehensive evaluation and comparison you've provided.

Table 7: The prompt template used for the proxy judge model trained on the AutoJ-pairwise dataset.

You are assessing two submitted responses to a given user's query. Your task is to **identify and articulate the flaws or weaknesses** in each response. These may include, but are not limited to: irrelevance, factual inaccuracies, logical fallacies, ambiguity, verbosity, or failure to address the core of the query.

[BEGIN DATA]
***
[Query]: {instruction}
***
[Response 1]: {response_A}
***
[Response 2]: {response_B}
***
[END DATA]

Please follow these instructions:

1. Critically analyze each response and **point out any notable issues, shortcomings, or limitations**.
2. For each response, **list its weaknesses in bullet points**, providing concise yet specific explanations.
3. If a response does not have major flaws, explicitly state that as well.

Focus on constructive and detailed critique — do not provide an overall preference or ranking between the two responses.

Table 8: The prompt for identifying the severe flaws of two candidate responses.

You are assessing two submitted responses on a given user's query and judging which response is better. Here is the data:

[BEGIN DATA]
***
[Query]: {instruction}
***
[Response 1]: {response_A}
***
[Response 2]: {response_B}
***
[END DATA]

Here are the instructions to assess and compare the two responses:

1. Pinpoint the key factors to distinguish these two responses.
2. Conclude your comparison by providing a final decision on which response is better. Begin your final decision statement with "So, the final decision is Response 1 / Response 2". Ensure that your decision aligns coherently with the comprehensive evaluation and comparison you've provided.

Below are two sample evaluations for the above comparison tasks. Use them as reference for structure, reasoning, and tone.

[Reference Evaluation 1]:
{evaluation_A}
[Reference Evaluation 2]:
{evaluation_B}

[Reference Evaluation End]

Now, based on the two evaluations, please provide your own evaluation for the tasks:

Table 9: The assistant model is prompted to provide a new evaluation based on evaluations from teacher model and assistant model directly.

You are assessing two submitted responses on a given user's query and judging which response is better. Here is the data:

[BEGIN DATA]
***
[Query]: {instruction}
***
[Response 1]: {response_A}
***
[Response 2]: {response_B}
***
[END DATA]

Here are the instructions to assess and compare the two responses:

1. Pinpoint the key factors to distinguish these two responses.
2. Conclude your comparison by providing a final decision on which response is better. Begin your final decision statement with "So, the final decision is Response 1 / Response 2". Ensure that your decision aligns coherently with the comprehensive evaluation and comparison you've provided.

Below is a reference evaluations for the above comparison tasks. Use it as reference for structure, reasoning, and tone.

[Reference Evaluation 1]:
{evaluation_A}

[Reference Evaluation End]

Now, based on the reference evaluation, please provide your own evaluation for the tasks:

Table 10: Rephrase the evaluation generated by the teacher model.

You are assessing two submitted responses on a given user's query and judging which response is better. Here is the data:

[BEGIN DATA]
***
[Query]: {instruction}
***
[Response 1]: {response_A}
***
[Response 2]: {response_B}
***
[END DATA]

Below are two sample evaluations for the above comparison task. Use them as reference for your own evaluation.

[Reference Evaluation 1 - Comparative Judgment]:
This evaluation determines **which of the two responses is better overall**, providing reasoning and a final decision.

{evaluation_A}

[Reference Evaluation 2 - Critical Analysis]:
This evaluation **identifies weaknesses or flaws in both responses**, such as irrelevance, logical errors, or failure to address the query effectively.

{evaluation_B}

[Reference Evaluation End]

Here are some rules of the evaluation:
1. Pinpoint the key factors to distinguish these two responses.
2. Ensure that your evaluation is self-contained and informative without needing to refer back to these examples.
3. Conclude your comparison by providing a final decision on which response is better, or they are tied. Begin your final decision statement with "So, the final decision is Response 1 / Response 2 / Tie". Ensure that your decision aligns coherently with the comprehensive evaluation and comparison you've provided.
Now, based on both evaluations, please provide your own evaluation for the task:

Table 11: Generate new evaluation by integrating the teacher model's feedback and the critical flaws identified by assistant model.

D Case Study

We provided specific judgment examples to illustrate the sources of teacher preference bias. Table 12 contains data from AutoJ-pairwise, and Table 13 is derived from the OffsetBias test set.

[Query]: Fact-check each of the claims in the following answer.
Question: If humanity we're to disappear, how long would it take the earth to return to a state of 'pre-humans'
Answer: It's impossible to accurately predict how quickly life would recover without humans, but it would be certain that many years would have to pass before any resemblance of an earlier human state was attained. It is likely that any creature that replaced humans as the predominant species on the planet would have more prominence and propensity to dominate other creatures than humans ever could.
[Response 1]: "It's impossible to accurately predict how quickly life would recover without humans, but it would be certain that many years would have to pass before any resemblance of an earlier human state was attained." - Correct . No one knows how long it would take for life to recover without humans.
"It is likely that any creature that replaced humans as the predominant species on the planet would have more prominence and propensity to dominate other creatures than humans ever could." - Mostly correct . It's likely that a species will replace humans as the predominant species on the planet, but it's impossible to say how that new species might compare to humans in terms of prominence and propensity to dominate other creatures.
[Response 2]: "It's impossible to accurately predict how quickly life would recover without humans, but it would be certain that many years would have to pass before any resemblance of an earlier human state was attained." - Correct . It is likely that any creature that replaced humans as the predominant species on the planet would have more prominence and propensity to dominate other creatures than humans ever could." - Not quite correct. While it is true climate change, arguably humanity's most indelible impact on the planet, is irreversible , life could recover, albeit slowly, over millions of years . "It is likely that any creature that replaced humans as the predominant species on the planet would have more prominence and propensity to dominate other creatures than humans ever could." - Could not tell.
[Evaluation]:
1. The key factors to distinguish these two responses:
- information richness: Response 2 provides more detailed and rich information, such as the irreversible impact of climate change and the potential recovery of life over millions of years. Response 1, while accurate, does not provide as much detail or additional information.
- accuracy: Both responses are accurate in their verification of the claims. However, Response 2 provides a more detailed explanation, mentioning the irreversible impact of climate change and the potential recovery of life over millions of years. This additional information enhances the accuracy of the response.
- depth of analysis: Response 2 provides a deeper analysis of the claims, discussing the irreversible impact of climate change and the potential recovery of life over millions of years. Response 1 provides a more surface-level analysis, simply stating that the claims are correct or mostly correct without going into much detail.
- citation: Response 2 cites the impact of climate change and the potential recovery of life over millions of years, which adds credibility to the response. Response 1 does not provide any citations or references.
- clarity of explanation: Both responses are clear in their explanations, but Response 2 provides a more comprehensive explanation, which could be more helpful to the user.
2. The final decision:
So, the final decision is Response 2. While both responses are accurate and clear, Response 2 provides a more detailed and comprehensive analysis, cites credible information, and provides richer information, making it the better response.

Table 12: An example of a judgment made by GPT-4 in the AutoJ-pairwise training set.

[Query]: Suggest two potential topics for a persuasive essay.
[Response 1]:
1. The importance of implementing stricter gun control laws in order to reduce gun violence in the United States.
2. The benefits of adopting a plant-based diet for both personal health and environmental sustainability
[Response 2]: Two engaging themes for debate in a classroom setting could be:
1. **Technology and Society: "Should Schools Replace Traditional Education with Technology-Driven Education?"**

Pros: A tech-driven education can offer personalized learning experiences, instant access to vast resources, interactive learning, and prepare students for a digital future.

Cons: It can increase the educational divide due to unequal access to technology, diminish critical social interaction, and may not cater to all learning styles.

This debate can lead to discussions on equity, the role of teachers, the balance between digital and face-to-face learning, and the impact of technology on cognitive and social development.
2. **Environmental Ethics: "Is It Morally Acceptable for Developed Countries to Limit the Industrial Growth of Developing Countries in the Interest of Climate Change?"**

Pros: Developed countries have historically contributed the most to climate change and thus have a responsibility to lead in mitigation efforts. Limiting industrial growth in developing countries could be a way to prevent further global damage.

Cons: Developing countries may argue that they have a right to industrialize and improve their citizens' standards of living just as developed countries have done. They may see restrictions as a continuation of colonial-era injustices and a hindrance to their economic sovereignty.

This theme is great for debating ethical considerations, global justice, the balance between economic development and environmental protection, and the responsibilities developed and developing countries have to one another and to the planet.
These two topics are complex and multifaceted, which can elicit profound and sophisticated engagement from students, fostering critical thinking, research skills, and empathy as they consider different sides of the arguments.
[Evaluation]:
1. The key factors to distinguish these two responses:

- additional information: Both responses provide additional information about the topics, but Response 2 provides more detailed and nuanced information, explaining the pros and cons of each topic.

- vividness: Response 2 uses vivid and descriptive language to describe the topics, while Response 1's language is more matter-of-fact.

- clarity: Both responses are clear in their communication, but Response 2's detailed explanation of the pros and cons of each topic helps clarify the complexity of the topics.

- completeness of instruction following: Response 2 follows the user's query more closely by suggesting two specific topics for a persuasive essay, while Response 1 suggests two topics without further elaboration.

- relevance: Both responses suggest relevant topics, but Response 2 provides more context and relevance by explaining the pros and cons of each topic.

- depth of knowledge: Response 2 demonstrates a deeper understanding of the topics and their implications, while Response 1's suggestion is more superficial.

- text quality: Both responses are grammatically correct and free of spelling errors, but Response 2's text is more fluent and coherent.

2. The final decision:

So, the final decision is Response 2. While both responses meet some of the criteria, Response 2 provides a more detailed and nuanced explanation of the topics, demonstrates a deeper understanding of the topics, follows the user's query more closely, and uses more vivid and descriptive language.

Table 13: An example of a judgment made by GPT-4 in the OffsetBias test set.